FRONT PRE-TENSIONER SEAT BELT [TWO-STEP DEPLOYMENT CONTROL SYSTEM]

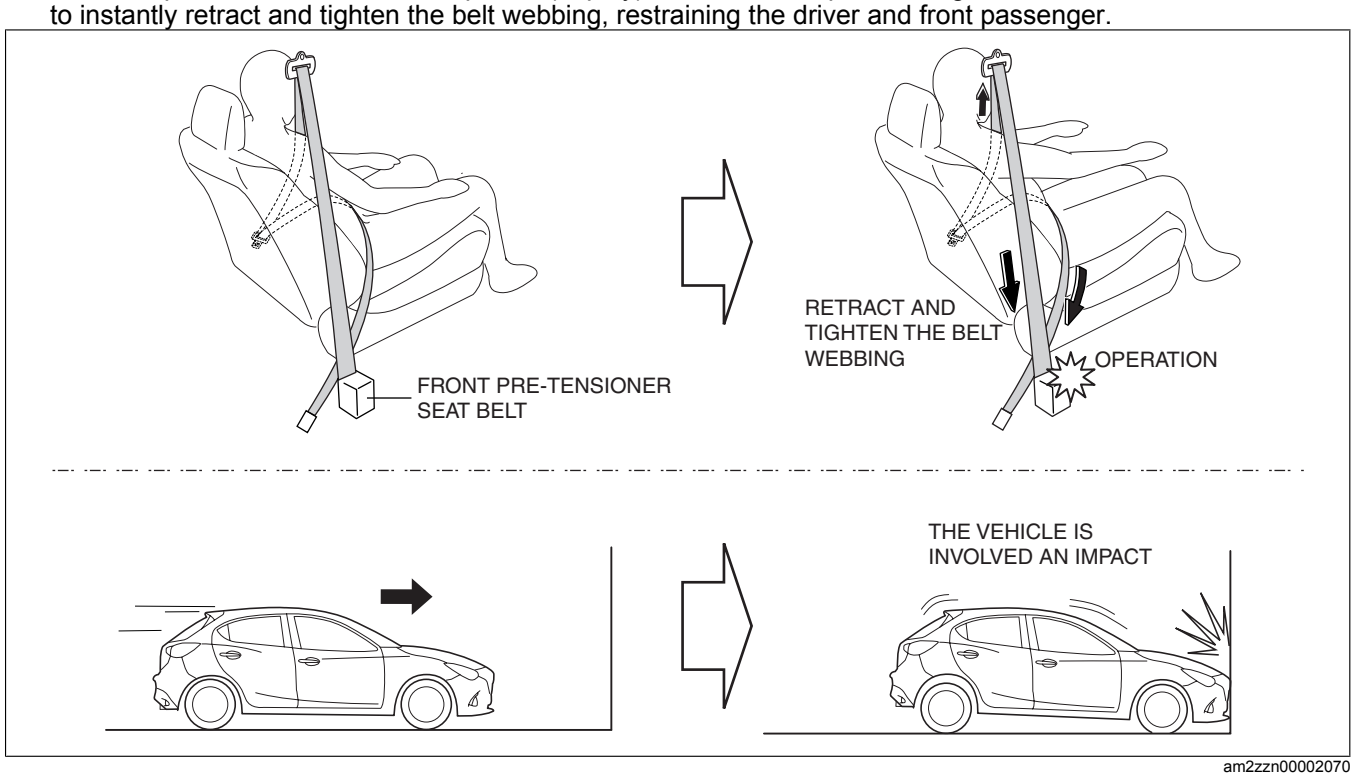
id0810b1502100

Purpose

- The front pre-tensioner seat belt retracts and tightens the belt webbing (between front buckle and pre-tensioner seat belt) against the occupant's upper torso to hold the occupant firmly in the seat and protect the occupant during a collision.

Function

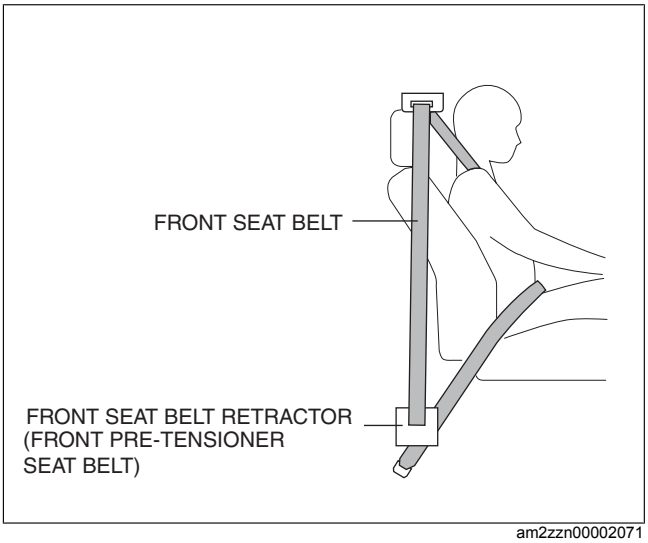
- The front pre-tensioner seat belts operate (deploy) based on the operation signal from the SAS control module to instantly retract and tighten the belt webbing, restraining the driver and front passenger.



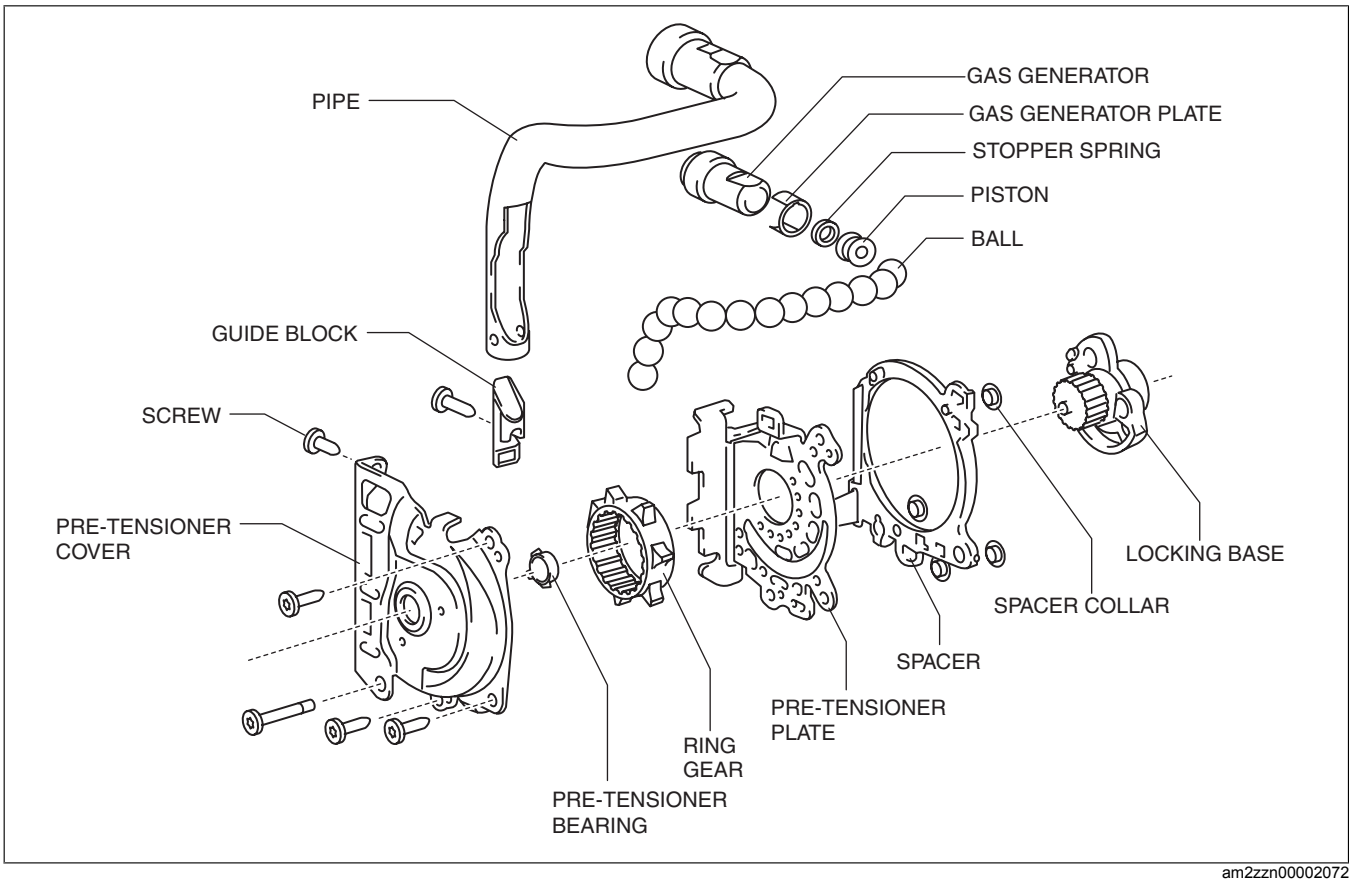
am2zzn00002070

Construction

- The front pre-tensioner seat belt is located in the retractor of the front seat belt.
- The front pre-tensioner seat belts consist of the following parts shown in the figure:



am2zzn00002071

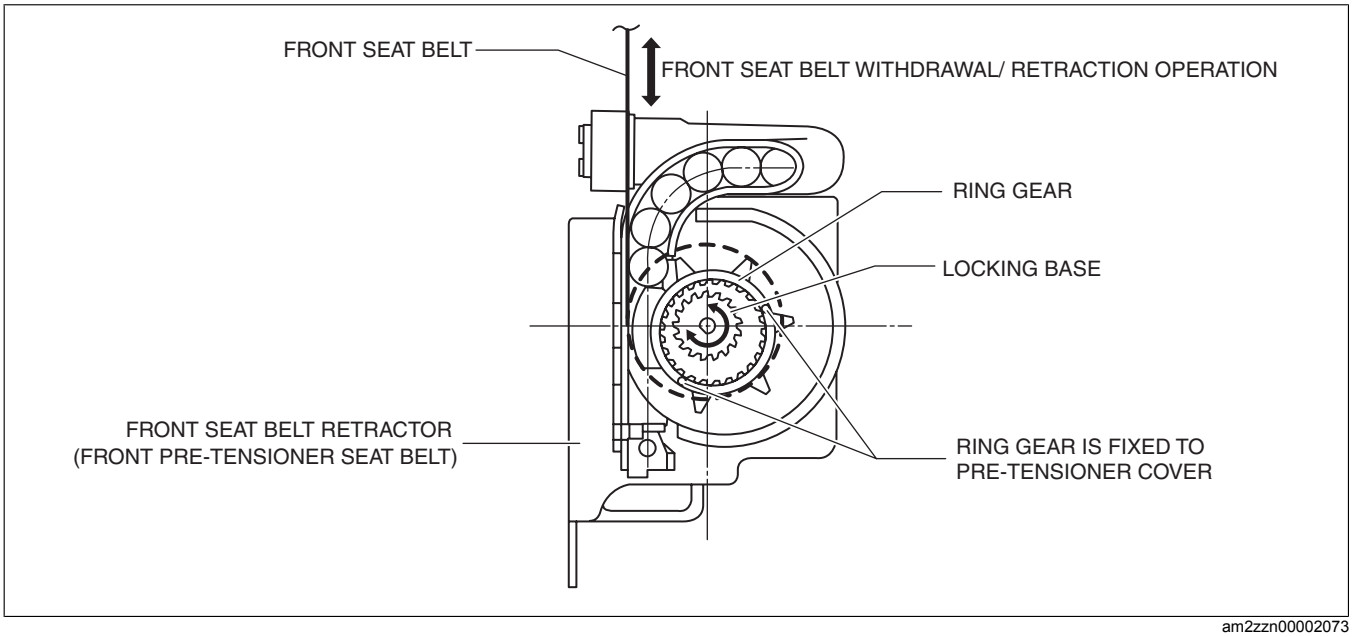


am2zzn00002072

Operation

Normal operation (pre-tensioner non-operation)

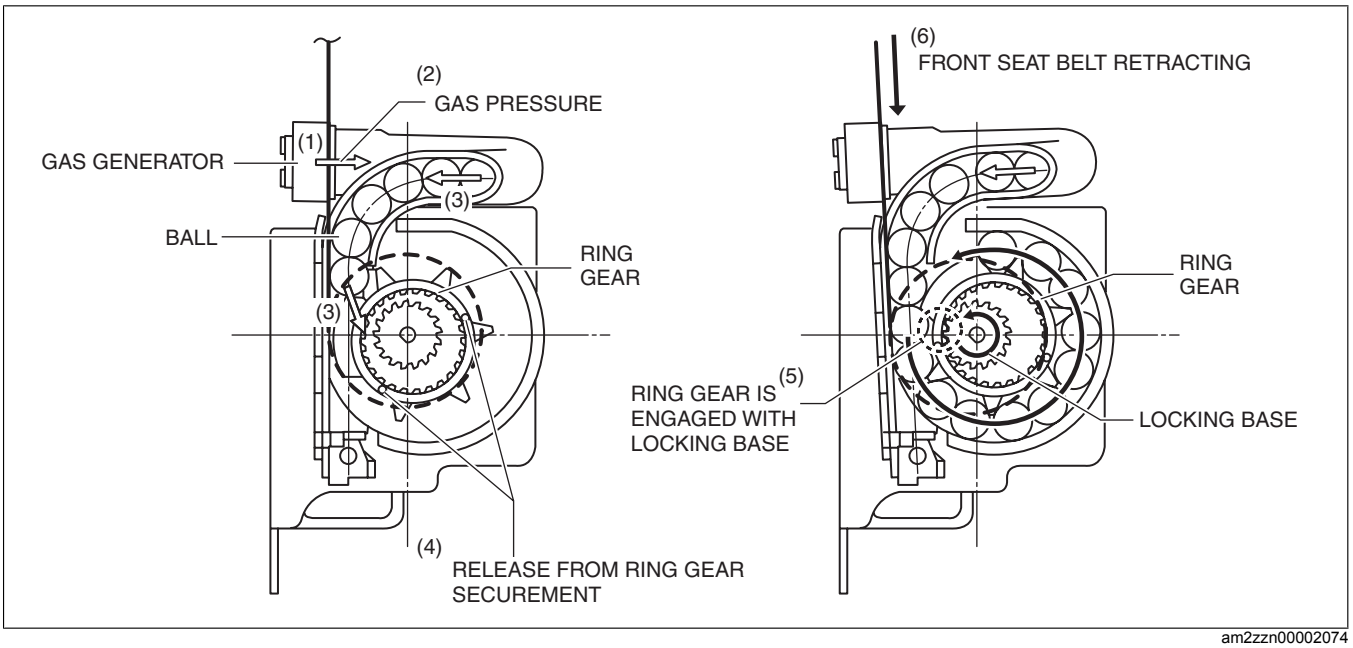
- The locking base of the front pre-tensioner seat belt is separated from the ring gear.
- The locking base of the front pre-tensioner seat belt rotates in conjunction with the withdrawl/retraction of the front seat belt.



am2zzn00002073

Pre-tensioner activation

1. The gas generator produces (1) gas when it receives an operation signal from the SAS control module.
2. The ball in the cylinder is pressed (3) by the generated gas pressure (2) and the ring gear is removed (4) from its securement.
3. The ring gear which is removed from its securement is caught in (5) the locking base, and the force of the pressed ball is transmitted to the locking base.
4. The locking base rotates and the front seat belt is retracted (6).



am2zzn00002074

Fail-safe

- Not applicable